Assignment 2

20/03/2026

Part 1

Exercise 1.1

We start by estimating the model.

```
 Intercept  Education Experience
     1.037      0.118      0.022
```

As the estimated OLS coefficients are small, we can conclude that one unit change in a regressor $x_{k,i}$ translates to $\beta_k \times 100\%$ change in y_i. In other words, each additional unit of education increases wages by 11.8%, whereas each additional unit of experience increases wages only by 2.2%.

Exercise 1.2

The correlation coefficient can be measured as

$$r_{\hat{\beta}_2,\hat{\beta}_3} = \frac{Cov(\hat{\beta}_2, \hat{\beta}_3)}{\sqrt{Var(\hat{\beta}_2)Var(\hat{\beta}_3)}}.$$

Computing the expression in R yields the following result.

```
[1] -0.056
```

Exercise 1.3

Part a)

```
t-test critical values at different significance levels:

at 1%  | critical value: 2.5858

at 5%  | critical value: 1.9648

at 10% | critical value: 1.6479

Regular test: t-test value: 1.2425  | p-value: 0.21463

Robust test: t-test value: 1.1707  | p-value: 0.2423
```

Part b)

```
Wald-test critical values at different significance levels (df=3):

at 1%  | critical value: 11.3449

at 5%  | critical value: 7.8147

at 10% | critical value: 6.2514

Regular test: Wald value: 3.3901  | p-value: 0.3353

Robust test:  Wald value: 5.4085  | p-value: 0.14422
```

Exercise 1.4

```
Breusch-Pagan Test (LM Statistic):

at 1%  | critical value: 9.2103

at 5%  | critical value: 5.9915

at 10% | critical value: 4.6052

Breusch-Pagan Test Results:

LM statistic:  13.6373  | p-value: 0.00109
```

Part 2

In the chunk below, a matrix is constructed containing a column with ones and two regressors taken from the given data files.

Exercise 2.1

One sample of $n = 500$

```
#set.seed(100)
#eps      = rnorm(n, 0, sigm0)                      # generate disturbances N(0, sigma^2)
#ysim     = Xdgp %*% beta0 + eps                    # together with X, your data set is complete

#
#  Code for exercise 2.1
#

#X = cbind( rep(1,n), , , ,  )          # setup the model you want to estimate
#k = ncol(X)

#XtX   = t(X) %*% X                     # matrix with basic statistics (n, sum(x), sum(x^2))
#XXinv = solve(XtX)                     # matrix used in Variance of estimator b
#diagXXinv = diag(XXinv)                # the diagonal elements (for the variance of b)
#XXinvX    = XXinv %*% t(X)             # part of the estimator betahat that remains the same
```

```
#
#  More code for exercise 2.1 (RESET test)
#
```

```
#
#  Code for exercise 2.2
#



# for (i in 1:nsim) {





#
#  More code for exercise 2.2
#


# }
```

For the next questions, copy/paste the code from the previous exercise and make the required adjustments.